BUDGET & BUDGETARY CONTROL

1. Budget – A budget is an instrument of management used as an aid in the planning, programming and control of business activity. In other words, budget is a blue-print of the projected plan of action expressed in quantitative terms for a specified period of time.

2. Budgetary Control – It is the system of management control and accounting in which all the operations are forecasted and planned in advance to the extent possible and the actual results compared with the forecasted and planned results.

Specification of Objectives → Setting goals and targets → Preparation of Budgets → Monitoring the actual results → Identifying the deviations → Taking preventive and corrective actions

3. Types of budget

Bud.
Sales (A) ✓
VC ✓
FC ✓
SVC ✓
TC (B) ✓
Pft. (A - B) ✓

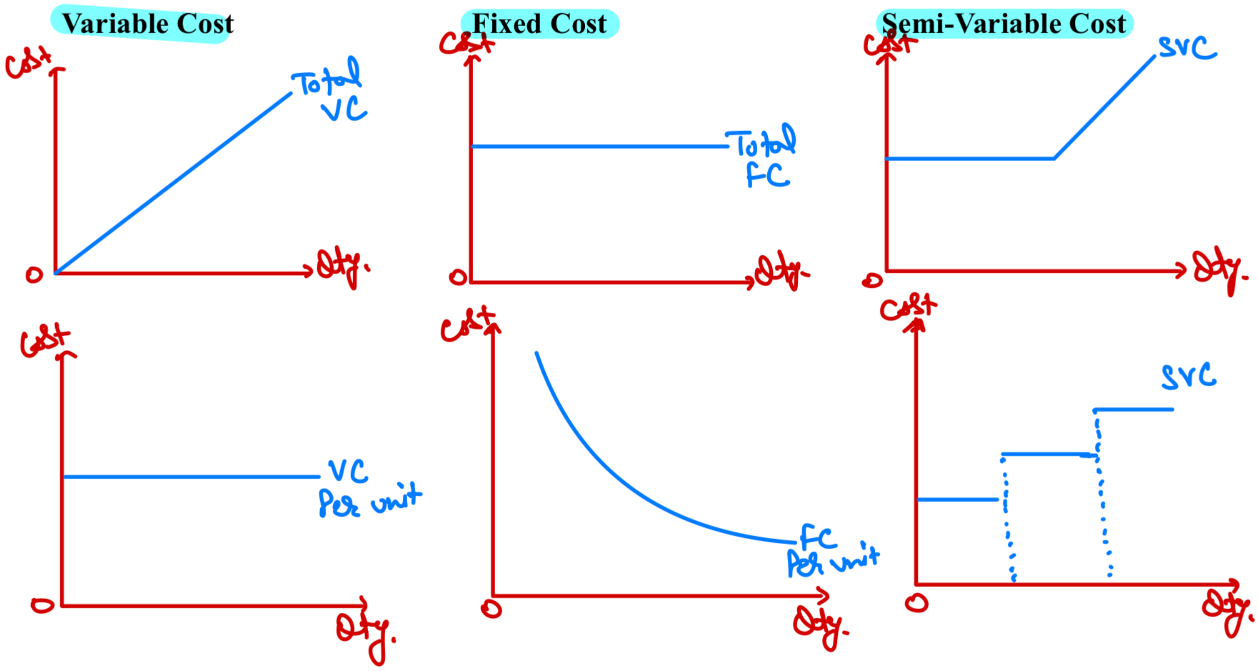
BUDGET

- 1. Capacity Wise
  - Fixed Budget
  - Flexible Budget
- 2. Function Wise
  - Sales Budget ✓
  - Production Budget ✓
  - Plant utilisation Budget
  - Direct-material usage Budget ✓
  - Direct-material purchase Budget ✓
  - Direct-labour (personnel) Budget ✓
  - Factory overhead Budget ✓
  - Production cost Budget
  - Ending inventory Budget
  - Cost of gooods-sold Budget
  - Selling and distribution cost Budget
  - Administration expenses Budget
  - Research and development cost Budget
  - Capital expenditure Budget
  - Cash Budget → in FM – Cash Mgt.
- Master Budget → Sales/Cost/Pft
- Period Wise
  - Long-term Budget → More than 5 Yrs.
  - Short-term Budget → 1 to 5 Yrs.
  - Current Budget → upto 1 Yr.

4. Fixed Budget – It is a budget designed to remain unchanged irrespective to the level of activity actually attained.

5. Flexible Budget – It is a budget which by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained.

6. Type of Costs



7. Total Cost = No. of units × Cost per unit

Quantity Effect   Price Effect

8. Quantity & Price Effect

	Total Variable Cost	Total Fixed Cost
Quantity Effect	**Yes**	**No**
Price Effect	**Yes**	**Yes**

9. Points to Remember (PTRs)

Unless otherwise provided, following assumptions are to be taken

- VC per unit will remain same
- Total Fixed cost will remain same
- All direct cost are assumed to be variable
- All overheads are assumed to be fixed

$\boxed{\text{Wages} \propto \frac{1}{Efficiency}}$ and $\boxed{\text{Efficiency} \propto \text{Output}}$

10. Distribution of Semi-Variable Cost

Variable cost per unit out of SVC = $\frac{Difference\ in\ cost}{Difference\ in\ units}$

Fixed cost out of SVC = Total cost – Total variable cost

Question – 1

PJ Ltd. manufactures hockey sticks. It sells the products at ₹ 500 each and makes a profit of ₹ 125 on each stick. The Company is producing 5,000 stocks annually by using 50% of its machinery capacity. The cost of each stick is as under:

Direct material V	₹ 150
Direct wages V	₹ 50
Work Overheads	₹ 125 (50% fixed)
Selling Expenses	₹ 50 (25% variable)

Fix. = 125×50% ×5000 = 312500
Var. = 125×50% = 62.5 P.u.
Fix : 50×75%×5000 = 187500
Var. = 50×25% = 12.50 Pu

The anticipation for the next year is that cost will go up as under:

Fixed charges → All Fix	10%
Direct wages	20%
Direct material	5%

There will not be any change in selling price There is an additional order for 2,000 sticks in the next year. Calculate the lowest price that can be quoted so that the Company can earn the same profit as it earned in the current year?

Total Prft

Solution

Statement of calculation of selling price

Particulars	Amount (₹)
Direct Material [(150 + 5%) × 7,000]	11,02,500
Direct Wages [(50 + 20%) × 7,000]	4,20,000
Variable Works Overhead [125 × 50% × 7,000]	4,37,500
Fixed Works Overhead [125 × 50% × 5,000 × 110%]	3,43,750
Variable Selling Expenses [50 × 25% × 7,000]	87,500
Fixed Selling Expenses [50 × 75% × 5,000 × 110%]	2,06,250
Total Cost	25,97,500
Add: Desired Profit (125 × 5,000)	625000 ~~8,75,000~~
Total Sales Value	~~34,72,500~~ 3222500
Less: Existing Sales from 5,000 units [5,000 × 500]	25,00,000
Sales value to be obtained from remaining 2,000 units (A)	~~9,72,500~~ 722500
Sale units (B)	2,000
Selling price per unit (A ÷ B)	~~486.25~~ 361.25

Question – 2

During the FY 2020-21, SK Limited has produced 60,000 units operating at 50% capacity level. The cost structure at the 50% level of activity is as under:

	(₹)
Direct material (V)	300 per unit
Direct wages (V)	100 per unit
Variable overheads (V)	100 per unit
Direct Expenses (V)	60 per unit
Factory expenses (25% fixed)	80 per unit
Selling and Distribution expenses (80% variable)	40 per unit
Office and administrative expenses (100 % fixed)	20 per unit

F = 80×25%×60000 = 12l
V = 80×75% = 60 p.u.

The company anticipates that in FY 2021-22, the variable costs will go up by 20% and fixed costs will go up by 15%. The selling price per unit will increase by 10% to ₹ 880. Required:

Original SP = 880/110% = 800

(i) Calculate the budgeted profit/loss for the FY 2020-21.

(ii) Prepare an expense budget on marginal cost basis for the FY 2021-22 for the company at 50% and 60% level of activity and find out the profits at respective levels.

Solution

(i) Calculation of Budgeted Profit for the year FY 2020-21

60% [60000/50% × 60% = 72000]

72000×880 =

50% FY 21-22

	60,000 Units	
	Per Unit	**Total**
Sales (A)	880/110% = 800.00	4,80,00,000 +10% = 528l
Variable Cost		
Direct material	300.00	1,80,00,000
Direct wages	100.00	60,00,000
Variable overheads	100.00	60,00,000
Direct expenses	60.00	36,00,000
Variable factory exp. (80×75%)	60.00	36,00,000
Variable selling exp. (40×80%)	32.00	19,20,000
Total Variable cost (B)	652	3,91,20,000 +20% = 46944000
Fixed Cost		
Office and admin. Exp. (100%) 20×60000 =	-	12,00,000
Fixed factory exp. (25%) 80×25%×60000 =	-	12,00,000
Fixed selling & dist. Exp. (20%) 40×20%×60000 =	-	4,80,000
Total Fixed cost (C)	-	28,80,000 +15% = 33.12l
Total cost (B+C = D)	-	4,20,00,000
Profit (A – D)	**-**	**60,00,000**

(652+20%) × 72000 =

28.80l +15% = 33.12

(ii) Expenses Budget for the year FY 2021-22 at 50% & 60% level

	60,000 units		72,000 units	
	Per Unit	**Total**	**Per Unit**	**Total**
Sales (A)	880	5,28,00,000	880	6,33,60,000

Variable Cost				
Direct material	360.00	2,16,00,000	360.00	2,59,20,000
Direct wages	120.00	72,00,000	120.00	86,40,000
Variable overheads	120.00	72,00,000	120.00	86,40,000
Direct expenses	72.00	43,20,000	72.00	51,84,000
Variable factory exp. (80×75%)	72.00	43,20,000	72.00	51,84,000
Variable selling exp. (40×80%)	38.40	23,04,000	38.40	27,64,800
Total Variable cost (B)	782.40	4,69,44,000	782.40	5,63,32,800
Fixed Cost				
Office and admin. Exp. (100%)	-	13,80,000	-	13,80,000
Fixed factory exp. (25%)	-	13,80,000	-	13,80,000
Fixed selling & dist. Exp. (20%)	-	5,52,000	-	5,52,000
Total Fixed cost (C)	-	33,12,000	-	33,12,000
Total cost (B + C = D)	-	5,02,56,000	-	5,96,44,800
Profit (A – D)	**-**	**25,44,000**	**-**	**37,15,200**

Question – 3

The Accountant of KPMR Ltd. has prepared the following budget for the coming year 2022 for its two products ‘AYE’ and ‘ZYE’:

Particulars	Product ‘AYE’	Product ‘ZYE’
Production and Sales (in Units)	4,000	3,000
	Amount (in ₹)	**Amount (in ₹)**
Selling price per unit	200	180
Direct material per unit	80	70
Direct labour per unit	40	35
Variable overhead per unit	20	25
Fixed overhead per unit	10 ×4000 =	10 ×3000 =

After reviewing the above budget, the management has called the marketing team for suggesting some measures for increasing the sales. The marketing team has suggested that by promoting the products on social media, the sales quantity of both the products can be increased by 5%. Also, the selling price per unit will go up by 10%. But this will result in increase in expenditure on variable overhead and fixed overhead by 20% and 5% respectively for both the products.

You are required to prepare flexible budget for both the products:

(i) Before promotion on social media

(ii) After promotion on social media

Solution

(i) Flexible Budget (Before promotion)

Particulars	Product AYE	Product ZYE	Total
Sales	4,000 × 200 = 8,00,000	3,000 × 180 = 5,40,000	13,40,000

Less: Direct Material	4,000 × 80 = 2,40,000	3,000 × 70 = 2,10,000	4,50,000
Less: Direct labour	4,000 × 40 = 1,60,000	3,000 × 35 = 1,05,000	2,65,000
Less: Variable OHs	4,000 × 20 = 80,000	3,000 × 25 = 75,000	1,55,000
Less: Fixed OHs	4,000 × 10 = 40,000	3,000 × 10 = 30,000	70,000
Profit	2,80,000	1,20,000	4,00,000

4000+5% = 4200 | 200+10% = 220 | 3000+5% = 3150 | 180+10% = 198

(ii) Flexible Budget (After promotion)

Particulars	Product AYE	Product ZYE	Total
Sales	4,200 × 220 = 9,24,000	3,150 × 198 = 6,23,700	15,47,700
Less: Direct Material	4,200 × 80 = 3,36,000	3,150 × 70 = 2,20,500	5,56,500
Less: Direct labour 20+20%	4,200 × 40 = 1,68,000	3,150 × 35 = 1,05,000	2,73,000
Less: Variable OHs	4,200 × 24 = 1,00,800	3,150 × 25 = ~~1,10,250~~ 94500 (25+20% = 30)	2,11,050
Less: Fixed OHs	40,000 + 5% = 42,000	30,000 + 5% = 31,500	73,500
Profit	2,77,200	1,56,450	4,33,650

Question – 4

SK Ltd. normally produce 8,000 units of their product in a month, in their machine shop. For the month of January, they had planned for a production of 10,000 units. Owing to a sudden cancellation of a contract in the middle of January, they could only produce 6,000 units in January.

Indirect manufacturing costs are carefully planned and monitored in the machine shop and the foreman of the shop is paid a 10% of the savings as bonus when in any month the indirect manufacturing cost incurred is less than the budgeted provision.

The foreman has put in a claim that he should be paid a bonus of ₹88.50 for the month of January. The works manager wonders how anyone can claim a bonus when the company has lost a sizeable contract. The relevant figures are as under:

Indirect manufacturing	Expenses for a normal month (₹) (8000 units)	Planned for January (₹) (10000 units)	Actual in costs January (₹) (6000 units)	6000 units Revised Plan
Salary of foreman	1,000	1,000	1,000	1000
Indirect labour (V) $\frac{720}{8000} \times 6000$	720	900	600	540
Indirect material (V) $\frac{800}{8000} \times 6000$	800	1,000	700	600
Repairs and maintenance (SVC)	600	650	600	550
Power (SVC)	800	875	740	725
Tools consumed (V) $\frac{320}{8000} \times 6000$	320	400	300	240
Rates and taxes	150	150	150	150
Depreciation	800	800	800	800
Insurance	100	100	100	100
	5,290	5,875	4,990	4705

$VC = \frac{650-600}{2000} = 0.025$

$FC = 650 - (10000 \times 0.025) = 400$

$(6000 \times 0.025) + 400 = 550$

$VC = \frac{875-800}{2000} = 0.0375$

$Fix. = 800 - (8000 \times 0.0375) = 500$

$(6000 \times 0.0375) + 500 = 725$

Inc. = 285

No Bonus

Do you agree with the works manager? Is the foreman entitled to any bonus for the performance in January? Substantiate your Answer with facts and figures. Explain.

Solution

Flexible Budget (for the month of January)

Indirect manufacturing	Nature of cost	Expenses for a normal month	Planned expenses for January	Expenses as per flexible budget for the month of January	Actual expenses of the month of January	Difference for Increased or (Decreased)
	(1)	(₹) (2)	(₹) (3)	(₹) (4)	(₹) (5)	(₹) (6 = 5 - 4)
Salary of foreman	Fixed	1,000	1,000	1,000	1,000	Nil
Indirect labour (Refer to working note 1)	Variable	720	900	540	600	60
Indirect material (Refer to working note 2)	Variable	800	1,000	600	700	100
Repairs and maintenance (Refer to working note 3)	Semi-variable	600	650	550	600	50
Power (Refer to working note 4)	Semi-variable	800	875	725	740	15
Tools consumed (Refer to working note 5)	Variable	320	400	240	300	60
Rates and taxes	Fixed	150	150	150	150	Nil
Depreciation	Fixed	800	800	800	800	Nil
Insurance	Fixed	100	100	100	100	Nil
Total		5,290	5,875	4,705	4,990	285

Conclusion: The above statement of flexible budget clearly shows that the concern's expenses in the month of January have increased from ₹4,705 to ₹4,990. Under such circumstances the Foreman of the company is not at all entitled for any performance bonus in January.

Working Notes:

1. Indirect labour cost per unit = $\frac{720}{8,000}$ = ₹0.09

   Indirect labour for 6,000 units = 6,000 × 0.09 = ₹540
2. Indirect material cost per unit = $\frac{800}{8,000}$ = ₹0.10

   Indirect material for 6,000 units = 6,000 × 0.10 = ₹600
3. According to high and low point method of segregating semi-variable cost into fixed and variable components, following formula may be used.

   Variable cost of repair and maintenance per unit = $\frac{\textit{Change in expense level}}{\textit{Change in Output level}} = \frac{650-600}{2,000}$ = ₹0.025

For 8,000 units: Total variable cost of repair & maintenance = 8,000 × 0.025 = ₹200

Fixed repair & maintenance cost = ₹600 – 200 = ₹400

For 6,000 units: Total cost of repair & maintenance = (6,000 × 0.025) + 400 = ₹550

4. Variable cost of power per unit = $\frac{875-800}{2,000}$ = ₹0.0375

For 8,000 units: Total variable cost of power = 8,000 × 0.0375 = ₹300

Fixed repair & maintenance cost = ₹800 – 300 = ₹500

For 6,000 units: Total cost of repair & maintenance = (6,000 × 0.0375) + 500 = ₹725

5. Tools consumed cost for 6,000 units = $\frac{320}{8,000}$ × 6,000 = ₹240

Question – 5

The cost accountant of manufacturing company provides you the following details for year 2018:

V.OH = $\frac{10}{10}$ ×100 = 100% of DW

	₹		₹
Direct materials	1,75,000	Other variable cost	80,000
Direct wages	1,00,000	Other fixed costs	80,000
Fixed factory overheads	1,00,000	Profit	1,15,000
Variable factory overheads	1,00,000	Sales	7,50,000

During the year, the company manufactured two products A and B and the output and costs were:

	A	B
Output (units)	2,00,000 -25% = 1.50L	1,00,000 -50% = 0.50L
Selling price per unit	₹2.00	₹3.50
Direct material per unit	₹0.50	₹0.75
Direct wages per unit	₹0.25	₹0.50

Variable factory overheads are absorbed as a percentage of direct wages. Other variable costs have been computed as: Product A ₹0.25 per unit; and B ₹0.30 per unit.

During 2019, it is expected that the demand for product A will fall by 25% and for B by 50%. It is decided to manufacture a further product C, the cost for which are estimated as follows:

	Product C
Output (units)	2,00,000
Selling price per unit	₹1.75
Direct materials per unit	₹0.40
Direct wages per unit	₹0.25

It is anticipated that the other variable costs 0.25 per unit will be the same as for product A.

Prepare a budget to present to the management, showing the current position and the position for 2019. Comment on the comparative results.

Solution

Budget Showing current Position and Position for 2019

Particulars	Current Position			Position for 2019			
	A	B (A+B)	Total	A	B	C ~~(A+B+C)~~	Total (A+B+C)
Sales (units)	2,00,000	1,00,000	-	1,50,000	50,000	2,00,000	-
	₹	₹	₹	₹	₹	₹	₹
A. **Sales** (₹)	4,00,000	3,50,000	7,50,000	3,00,000	1,75,000	3,50,000	8,25,000

Particulars	Current Position			Position for 2019			
Direct materials	1,00,000	75,000	1,75,000	75,000	37,500	80,000	1,92,500
Direct wages	50,000	50,000	1,00,000	37,500	25,000	50,000	1,12,500
Factory overhead	50,000	50,000	1,00,000	37,500	25,000	50,000	1,12,500
(var) (100% of DW)	50,000	30,000	80,000	37,500	15,000	50,000	1,02,500
Other variable costs	2,50,000	2,05,000	4,55,000	1,87,500	10,25,000	2,30,000	5,20,000
B. **Variable cost**	1,50,000	1,45,000	2,95,000	1,12,5000	72,500	1,20,000	3,05,000
C. **Contribution** (A-B)			1,00,000				1,00,000
Fixed cost: Factory			80,000				80,0000
Other			1,80,000				1,80,000
D. Total fixed cost			1,15,000				1,25,000
Profit (C – D)							

Comments: Introduction of product C is likely to increase profit by ₹10,000 (i.e. from ₹1,15,000 to ₹1,25,000) in 2019 as compared to 2018 Therefore, introduction of product C is recommended.

11. Sales Budget

Product	Units	Selling price per unit	Sales
A	-	-	-
B	-	-	-
C	-	-	-
Total	-		-

12. Production Budget

Sales = Op. St. + Prod. – Cl. St. ⇒ Prod. = Sales + Cl. St – Op. St.

Particulars	Product A	Product B	Product C
Sales	-	-	-
(+) Closing Stock	-	-	-
(-) Opening Stock	-	-	-
Production units	-	-	-

13. Raw Material Consumption Budget

Product	Prod. Quantity	RM consumption per unit	Total RM Consumption
A	-	-	-
B	-	-	-
C	-	-	-
	-	-	-

14. Raw Material Purchase Budget

RM Cons. = Op. RM + Purch. – Cl. RM

RM Purch. = RM Cons. + Cl. RM – Op. RM

Particulars	January	February	March
Raw Material Consumed	-	-	-
(+) Closing Stock	-	-	-
(-) Opening Stock	-	-	-
Raw Material Purchased	-	-	-

15. Labour Budget

Particulars	Product A	Product B	Product C
Production Units	-	-	-
Direct labour hour per unit	-	-	-
Total Labour hours	-	-	-
Total Labour cost @ Rs. __ per hour	-	-	-

Question – 6

PSV Ltd. manufactures and sells a single product and estimated the following related information for the period November, 2020 to March, 2021.

Prod 25500 41000 37000 23000 44000

(−) Opening Stock; (+) Cl. St. 3000 9000 8000 6000 10000

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Opening Stock of Finished goods (in Units)	7,500	3,000	9,000	8,000	6,000
Sales (in Units)	30,000	35,000	38,000	25,000	40,000
Selling Price per unit (in ₹)	10	12	15	15	20

Additional information:

- Closing stock of finished goods at the end of march, 2021 is 10,000 units
- Each unit of finished output requires 2kg of Raw Material ‘A’ and 3kg of Raw Material ‘B’.

You are required to prepare the following budgets for the period November, 2020 to March 2021 on monthly basis:

(i) Sales budget (in ₹)

(ii) Production Budget (in units) and

(iii) Raw material budget for raw material ‘A’ and ‘B’ separately (in units)

Solution

(i) Sales Budget

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Sales (in Units)	30,000	35,000	38,000	25,000	40,000
Selling Price per unit (in ₹)	10	12	15	15	20
Sales Value	3,00,000	4,20,000	5,70,000	3,75,000	8,00,000

(ii) Production Budget

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Sales Units	30,000	35,000	38,000	25,000	40,000
Add: Closing Stock Units	3,000	9,000	8,000	6,000	10,000
Less: Opening Stock Units	(7,500)	(3,000)	(9,000)	(8,000)	(6,000)
Production Units	25,500	41,000	37,000	23,000	44,000

(iii) Raw Material 'A' Budget

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Production Units	25,500	41,000	37,000	23,000	44,000
Raw material consumption per unit	2	2	2	2	2
Raw Material Consumption	51,000	82,000	74,000	46,000	88,000

Raw Material 'B' Budget

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Production Units	25,500	41,000	37,000	23,000	44,000
Raw material consumption per unit	3	3	3	3	3
Raw Material Consumption	76,500	1,23,000	1,11,000	69,000	1,32,000

Question – 7

SK Ltd. is drawing a production plan for its two products Minimax (MM) and Heavyhigh (HH) for the year 2019-20. The company's policy is to hold closing stock of finished goods at 25% of the anticipated volume of sales of the succeeding month. The following are the estimated data for two products:

	Minimax (MM)	Heavyhigh (HH)
Budgeted Production units	1,80,000	1,20,000
	(₹)	**(₹)**
Direct material cost per unit	220	280
Direct labour cost per unit	130	120
Manufacturing overhead (F)	4,00,000	5,00,000

The estimated units to be sold in the first four months of the year 2019-20 are as under:

	April	May	June	July
Minimax	8,000	10,000	12,000	16,000
Heavyhigh	6,000	8,000	9,000	14,000

Prepare production budget for the first quarter in month-wise.

Solution

Production budget (in Units)

	April		May		June		Total	
	MM	HH	MM	HH	MM	HH	MM	HH
Sales	8,000	6,000	10,000	8,000	12,000	9,000	30,000	23,000
Add: Closing stock (25% of Sale of MM)	2,500	2,000	3,000	2,250	4,000	3,500	9,500	7,750
Less: Opening stock	(2,000)	(1,500)	(2,500)	(2,000)	(3,000)	(2,250)	(7,500)	(5,750)
Production units	8,500	6,500	10,500	8,250	13,000	10,250	32,000	25,000

Production Cost Budget

Element of cost	Rate (₹) MM (32,000 units)	Rate (₹) HH (25,000 units)	Amount (₹) MM	Amount (₹) HH
Direct Material	220	280	70,40,000	70,00,000
Direct Labour	130	120	41,60,000	30,00,000
Manufacturing Overhead				
[(4,00,000÷1,80,000)×32,000]			71,111	
[(5,00,000÷1,20,000)×25,000]				1,04,167
			1,12,71,111	1,01,04,167

Question – 8	

AB manufacturing Company manufactures two products A and B. Both Products use a common Raw Material 'C'. The Raw Material 'C' is purchased at the rate of ₹ 45 per kg from the Market. The Company has made estimates for the year ended 31st March, 2018 (the budget period) as under:

	Product A		Product B	
Sales in Units	36,000	36860	16,700	17100
Cl. - Op Finished goods stock increase by year-end (in Units)	860	÷97%	400	÷95%
Post-production Rejection Rate (%)	3	= 38000 ×4	5	= 18000 ×5
Material 'C' per completed Unit, net of wastage	4 kg	= 152000 ÷95%	5 kg	= 90000 ÷96%
Material 'C' wastage in %	5	= 1.60L	4	= 93750

253750

Additional information available is as under:

- Usage of Raw Material 'C' is expected to be at a constant rate over the period.
- Annual cost of holding one unit of Raw Material 'C' in Stock is 9% of the Material Cost.
- The cost of placing an order is ₹ 250 per order.

You are required to:

(i) Prepare Functional Budgets for the year ended 31st March, 2018 under the following categories:

  a) Production Budget for Products A and B in Units

  b) Purchase Budget for Raw Material 'C' in kg and value.

(ii) Calculate the Economic Order Quantity (EOQ) in kg for Raw Material 'C'.

$\sqrt{\frac{2 \times 253750 \times 250}{(9\% \times 45)}}$ = 5597

Solution

(i) **Production Budget (in units) for the year ended 31st March 2018**

Particulars	Product A	Product B
Budgeted sales (units) Add: Increase in closing stock	36,000 860	16,700 400
No. of good units to be produced	36,860	17,100
Post production rejection rate	3%	5%
Post production good units rate	100% – 3% = 97%	100% - 5% = 95%
No. of units to be produced	36,860 ÷ 97% = 38,000	17,100 ÷ 95% = 18,000

(ii) **Purchase budget (in kgs and value) for Material C**

Particulars	Product A	Product B
No. of units to be produced Usage of Material C per unit of production	38,000 4 kg	18,000 5 kg
Material needed for production	1,52,000 kg	90,000 kg
Wastage % of Material C	5%	4%
Good usage % of Material C	100% - 5% = 95%	100% - 4% = 96%
Material to be purchased (in kg) Rate per kg of Material C	1,52,000÷95% = 1,60,000 ₹ 45	90,000÷96% = 93,750 ₹ 45
Total Purchase cost	1,60,000×45 = 72,00,000	93,750×45 = 42,18,750

Total purchase cost = 72,00,000 + 42,18,750 = ₹ 1,14,18,750

(iii) A = 1,60,000 + 93,750 = 2,53,750 kg

O = ₹ 250

C = ₹ 45 × 9% = ₹ 4.05

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 2{,}53{,}750 \times 250}{4.05}} = 5{,}597 \text{ kg}$$

Question – 9	

A Limited has furnished the following information for the months from 1st January to 30th April, 2023:

	January	February	March	April
Number of working days	1250 25	1320 24	1560 26	1300 25
Production (in units) per working day	50	55	60	52
Raw material purchases (% by weights to total of 4 months)	20800 × 21% = 4368	20800 × 26% = 5408	20800 × 30% = 6240	20800 × 23% = 4784
Purchase price of raw material (per kg)	₹ 10 43680	₹ 12	₹ 13	₹ 11

Total
5430
× 4
= 21720
+ 5100
− 6020
20800

Quantity of raw material per unit of product: 4 kg

Opening stock of raw material on 1st January: 6,020 kg. (Cost ₹ 63,210)

Closing stock of raw material on 30th April: 5,100 kg.

All the purchases of material are made at the start of each month.

Required:

(i) Calculate the consumption of raw material (in kgs) month-by-month and in total

(ii) Calculate the month-wise quantity and value of raw materials purchased

(iii) Prepare the priced stores ledger for each month using the FIFO method.

Solution

(i) **Calculation of consumption for Raw Material (in kgs) month by month and total**

Particulars	Jan	Feb	March	April	Total
No. of working cays Production (per day)	25 50	24 55	26 60	25 52	- -
Production	1,250	1,320	1,560	1,300	5,430
Raw material consumed (in kgs)	5,000	5,280	6,240	5,200	21,720

Calculation of Raw Material Purchased

Purchased	Kg
Closing stock on 30th April	5,100
Add: Raw material consumed	21,720
Less: Opening stock on 1st January	(6,020)
Raw material purchased	20,800

(ii) **Calculation of month wise quantity and value of raw materials purchased**

Month	Purchase quantity (Kgs)	Price (₹)	Value (₹)
January	20,800 × 21% = 4,368	10	43,680
February	20,800 × 26% = 5,408	12	64,896
March	20,800 × 30% = 6,240	13	81,120
April	20,800 × 23% = 4,784	11	52,624
Total	20,800		2,42,320

(iii) **Stores Price Ledger by using FIFO Method**

Month	Receipts			Issues			Balance		
	Qty. (kg)	Rate (₹)	Amount	Qty. (kg)	Rate (₹)	Amount	Qty. (kg)	Rate (₹)	Amount
Jan	-	-	-	-	-	-	6,020	10.5	63,210
Jan	4,368	10	43,680	-	-	-	6,020 4,368	10.5 10	63,210 43,680
Jan	-	-	-	5,000	10.5	52,500	1,020 4,368	10.5 10	10,710 43,680
Feb	5,408	12	64,896	-	-	-	1,020 4,368 5,408	10.5 10 12	10,710 43,680 64,896
Feb	-	-	-	1,020 4,260 5280	10.5 10	10,710 42,600	108 5,408	10 12	1,080 64,896
March	6,240	13	81,120	-	-	-	108 5,408 6,240	10 12 13	1,080 64,896 81,120
March	-	-	-	108 5,408 724 6240	10 12 13	1,080 64,896 9,412	5,516	13	71,708
April	4,784	11	52,624	-	-	-	5,516 4,784	13 11	52,624 71,708
April	-	-	-	5,200	13	67,600	316 4,784	13 11	4,108 52,624

Question – 10

SK Ltd. Produces and sells a single product. Sales budget for the calendar year 2019 by quarter is as under:

Quarter	No. of units to be sold
I	12,000
II	15,000
III	16,500
IV	18,000
	61500

The year is expected to open with an inventory of 4,000 (op.) units of finished product and close with an inventory of 6,500 (CL.) units.

Production is customarily scheduled to provide for two-thirds of the current quarter's sales demand plus one third of the following quarter's demand. Thus, production anticipates sales volume by about one month.

The standard cost details for one unit of the product is as follows:

Direct materials 10 lbs. @ 50 paise per lb. → $10 \times 0.50 = 5$ P.u

Direct labour 1 hour 30 minutes @ ₹ 4 per hour. → $\frac{90}{60} \times 4 = 6$ P.u

Variable overheads 1 hour 30 minutes @ ₹ 1 per hour. → $\frac{90}{60} \times 1 = 1.5$ P.u

VC @ 12.5 P.u

Fixed overheads 1 hour 30 minutes @ ₹ 2 per hour based on a budgeted volume of 90,000 direct labour hour for the year.

Total F. OH = 2×90000 = 1.80L

(i) Prepare a production budget for 2019, by quarters, showing the number of units to be produced, and the total costs of direct labour, variable overheads and fixed overheads.

(ii) If the budgeted selling price per unit is ₹ 17, what would be the budgeted profit for the year as a whole?

Sale = 17×61500 = 10.455L
(−)VC = 12.5×61500 = 768750
(−)FC = 1.80L
Pft = 96750

(iii) In which quarter of the year is the company expected to breakeven?

$BEP = \frac{1.80L}{(17-12.5)} = 40,000$ units

Solution

(i) **Production Budget**

Particulars	Quarter - 1	Quarter – 2	Quarter – 3	Quarter – 4	Total
Sales units	12,000	15,000	16,500	18,000	61,500
Production					
2/3 of current month	8,000	10,000	11,000	12,000	41,000
1/3 of next month	5,000	5,500	6,000	6,500*	23,000
Production	**13,000**	**15,500**	**17,000**	**18,500**	**64,000**

*This value of 6,500 units is computed as balancing figure.

Working Note -

Annual total production = Sales + closing stock – Opening stock = 61,500 + 6,500 – 4,000 = 64,000

Production for Quarter - 4 = Total production – production of first three quarters

= 64,000 – 13,000 – 15,500 – 17,000 = 18,500

Statement of Cost

Particulars	**Quarter - 1**	**Quarter – 2**	**Quarter – 3**	**Quarter – 4**	**Total**
Production units	13,000	15,500	17,000	18,500	64,000
Direct material (10 × 0.5 = 5 p.u.)	65,000	77,500	85,000	92,500	3,20,000
Direct labour (1.5 × 4 = 6 p.u.)	78,000	93,000	1,02,000	1,11,000	3,84,000
Variable Ohs (1.5 × 1 = 1.5 p.u.)	19,500	23,250	25,500	27,750	96,000
Fixed overheads $\left(\frac{1.80L}{4}\right)$	45,000	45,000	45,000	45,000	1,80,000 (90,000×2)

(ii) **Statement of profit**

Particulars	**Amount**
Sales (17 × 61,500)	10,45,500
Less: Variable cost $\left[\frac{(3{,}20{,}000+3{,}84{,}000+96{,}000)}{64{,}000} \times 61{,}500\right]$	7,68,750
Contribution	2,76,750
Less: Fixed costs	1,80,000
Profit	96,750

(iii) Breakeven point $= \frac{\textit{Fixed Cost}}{\textit{Contribution per unit}} = \frac{1{,}80{,}000}{\left(\frac{2{,}76{,}750}{61{,}500}\right)} =$ 40,000 units

Particulars	**Quarter - 1**	**Quarter – 2**	**Quarter – 3**	**Quarter – 4**
Sales units	12,000	15,000	16,500	18,000
Cumulative Sales units	12,000	27,000	43,500	61,500

∴ Breakeven will be achieved in quarter 3.

Question – 11

SK Ltd. manufactures two products using two types of materials and one grade of labour. Shown below is an extract from the company's working papers for the next month's budget:

	Product-A	**Product-B**
Budgeted sales (in units)	2,400	3,600
Budgeted material consumption per unit (in kg):		
Material-X	5	3
Material-Y	4	6
Standard labour hours allowed per unit of product	3	5

Material-X and Material-Y cost ₹ 4 and ₹ 6 per kg and labours are paid ₹ 25 per hour. Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 180 direct workers.

	A	B
Sale	2400	3600
(+) Cl.	480	900
(–) op.	(400)	(200)
Prod.	2480	4300

	X	Y
Cons.	(2480×5)+(4300×3) = 25300	(2480×4) + (4300×6) = 35720
(+) Cl. St	12650	10716
(–) op. St	(1000)	(500)
Purch.	36950	45936

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%. In addition the non-productive down-time is budgeted at 20% of the productive hours worked.

4×5 = 20 days

There are four 5-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period.

Wages
Nor Hrs. (2480×3)+(4300×5) = 28940
Prod. Hrs. 28940 ÷ 80% = 36175
(+) 20% 7235
Hrs. Paid 43410
(−) Nr. Hrs. (180×40×4) 28800
OT Hrs 14610

It is anticipated that stock at the beginning of the period will be:

op
- Product-A = 400 units; Product-B = 200 units;
- Material-X = 1,000 kgs; Material-Y = 500 kgs.

The anticipated closing stocks for budget period are as below:

Q.
- Product-A 4 days sales → $\frac{2400}{20} \times 4 = 480$
- Product-B 5 days sales → $\frac{3600}{20} \times 5 = 900$
- Material-X 10 days consumption → $\frac{25300}{20} \times 10 = 12650$
- Material-Y 6 days consumption → $\frac{35720}{20} \times 6 = 10716$

Wages = (28800×25) + (14610×(25+50%)) = 1267875

Required to calculate the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

Solution

Number of days in budget period = 4 weeks × 5 days = 20 days

Number of units to be produced

	Product-A (units)	**Product-B (units)**
Budgeted Sales	2,400	3,600
Add: Closing stock $\left(\frac{2{,}400\ units}{20\ days} \times 4\ days\right)\left(\frac{3{,}600\ units}{20\ days} \times 5\ days\right)$	480	900
Less: Opening stock	400	200
	2,480	4,300

(i) Material purchase budget

	Material-X (kg)	**Material-Y (kg)**
Material required:		
Product-A	12,400 (2,480 units x 5 kg)	9,920 (2,480 units x 4 kg)
Product-B	12,900 (4,300units x 3 kg)	25,800 (4,300units x 6kg)
	25,300	35,720
Add: Closing Stock $\left(\frac{25{,}300\ kg}{20\ days} \times 10\ days\right)\left(\frac{35{,}720\ units}{20\ days} \times 6\ days\right)$	12,650	10,716
Less: Opening stock	1,000	500
Quantity to be purchased	36,950	45,936
Rate per kg. of material	₹ 4	₹ 6
Total cost	**₹ 1,47,800**	**₹ 2,75,616**

(ii) Wages Budget

	Product-A (Hours)	Product-B (Hours)
Units to be produced Standard hours allowed per unit	2,480 units 3	4,300 units 5
Total standard hours allowed	7,440	21,500
Productive hours required for production Add: Non-Productive down time	$\frac{7{,}440\ hours}{80\%}$ = 9,300 1,860 hours (20% of 9,300 hours)	$\frac{21{,}500\ hours}{80\%}$ = 26,875 5,375 hours (20% of 26,875 hours)
Hours to be paid	11,160	32,250

Total Hours to be paid = 43,410 hours (11,160 + 32,250)

Hours to be paid at normal rate = 4 weeks × 40 hours × 180 workers = 28,800 hours

Hours to be paid at premium rate = 43,410 hours – 28,800 hours = 14,610 hours

Total wages to be paid = (28,800 hours × ₹ 25) + (14,610 hours × ₹ 37.5)
= ₹ 7,20,000 + ₹ 5,47,875 = ₹ 12,67,875

16. Zero Base Budgeting – It is defined as 'a method of budgeting which requires each cost element to be specifically justified, although the activities to which the budget relates are being undertaken for the first time, without approval, the budget allowance is zero'.

ZBB is an activity based budgeting system where budgets are prepared for each activities rather than functional department. Justification in the form of cost benefits for the activity is required to be given. The activities are then evaluated and prioritized by the management on the basis of factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

17. Performance Budgeting – It involves evaluation of the performance of an organisation in the context of both specific as well as overall objectives of the organisation. Performance Budgeting provide a meaningful relationship between estimated inputs and expected outputs as an integral part of the budgeting system. A performance budget is one which presents the purposes and objectives for which funds are required, the costs of the programmes proposed for achieving those objectives, and quantitative data measuring the accomplishments and work performed under each programme. Thus, PB is a technique of presenting budgets for costs and revenues in terms of functions. Programmes and activities correlate the physical and financial aspect of the individual items comprising the budget.